3/7/2025 (FN)

Hall Ticket Number:

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MR-22

Code: B425CS2

# MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.Tech. IV Semester Regular/Supplementary Examinations, July 2025

## Discrete Mathematics

*(Common to CSE, IT and CSM)*

Time: 3 Hours | *Answer ALL Questions* | Max. Marks: 60

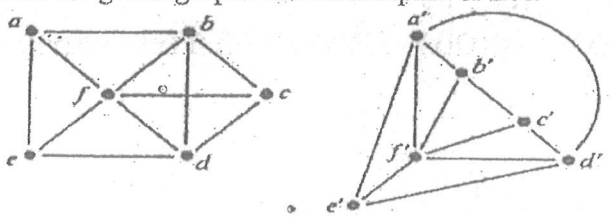
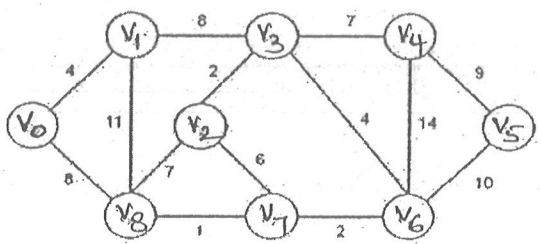
*Part-A (10 x 1 = 10 Marks)*

| Q. No. | Stem of the Question | M | L | CO | PO |
|---|---|---|---|---|---|
| 1. a) | When do we say that two statement formulas are Equivalent? | 1 | 2 | 1 | 1 |
| 1. b) | Define DNF. | 1 | 1 | 1 | 1 |
| 1. c) | What is Power set? | 1 | 1 | 2 | 2 |
| 1. d) | Give an example of a Injective function. | 1 | 2 | 2 | 2 |
| 1. e) | Define Monoid. | 1 | 1 | 3 | 1 |
| 1. f) | State any two properties of Lattices. | 1 | 2 | 3 | 2 |
| 1. g) | Define combination. | 1 | 1 | 4 | 1 |
| 1. h) | How many ways are there to distribute 10 different books among 15 people if no person is to receive more than 1book? | 1 | 2 | 4 | 3 |
| 1. i) | What is the degree of a graph? | 1 | 2 | 5 | 1 |
| 1. j) | What is the maximum number of edges in a simple undirected graph with n vertices? | 1 | 1 | 5 | 2 |

*Part-B (5 x 10 = 50 Marks)*

| Q. No. | Stem of the Question | M | L | CO | PO |
|---|---|---|---|---|---|
| 2. a) | Show that $P \rightarrow (Q \rightarrow R) \Leftrightarrow P \rightarrow (\neg Q \vee R) \Leftrightarrow (P \wedge Q) \rightarrow R$. | 5 | 2 | 1 | 2 |
| 2. b) | Verify whether the following premises are consistent or not.<br>1. If Jack misses many classes through illness, then he fails high school.<br>2. If Jack fails high school, then he is uneducated.<br>3. If Jack reads a lot of books, then he is not uneducated.<br>4. Jack misses many classes through illness and reads a lot of books. | 5 | 2 | 1 | 3 |
| | **OR** | | | | |
| 2. c) | Obtain PCNF for $(\neg P \rightarrow R) \wedge (Q \rightleftarrows P)$. | 5 | 2 | 1 | 3 |
| 2. d) | Show that $(x)(P(x) \rightarrow Q(x)) \wedge (x)(Q(x) \rightarrow R(x)) \Rightarrow (x)(P(x) \rightarrow R(x))$. | 5 | 2 | 2 | 3 |
| 3. a) | Show that for any two sets A and B, $A - (A \cap B) = A - B$. | 5 | 2 | 2 | 3 |
| 3. b) | State and explain about any four Primitive Recursive Functions. | 5 | 1 | 2 | 2 |
| | **OR** | | | | |
| 3. c) | Let $f(x)=x+2$, $g(x)=x-2$, $h(x)=3x$ for $x \in R$, where R is a set of real numbers. Find<br>(i) fog<br>(ii)gof,<br>(iii)fogoh,<br>(iv)gohof. | 5 | 3 | 2 | 3 |
| 3. d) | Show that the set of integers, positive, negative and zero is denumerable. | 5 | 3 | 2 | 1 |
| 4. a) | Discuss about Subsemigroups and Submonoids. | 5 | 1 | 3 | 2 |
| 4. b) | Let $A = \{1,2,3,6\}$ with partial order defined by divisibility (i.e., $a \leq b \Leftrightarrow a \mid b$). Prove that $(A, \leq)$ is a lattice by showing that LUB (join) and GLB (meet) exist for all pairs. | 5 | 3 | 3 | 3 |
| | **OR** | | | | |

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|---|---|---|---|---|---|
| 4. c) | Define an algebraic system. Give two examples of algebraic systems one with a single binary operation and another with two binary operations. | 5 | 2 | 3 | 2 |
| 4. d) | Summarize about Boolean Algebra and its properties. | 5 | 1 | 3 | 2 |
| 5. a) | There are 30 females and 35 males in the junior class while there are 25 females and 20 males in the senior class. In how many ways can a committee of 10 be chosen so that there are exactly 5 females and 3 juniors on the committee? | 5 | 3 | 4 | 3 |
| 5. b) | (i) In $(2x-3y+5z)^8$ find the coefficient of $x_1^3 x_2^3 x_3^2$.<br>(ii) In $(x_1+x_2+x_3+x_4+x_5)^{10}$ find the coefficient of $x_1^2 x_3 x_4^3 x_5^4$. | 5 | 2 | 4 | 3 |
| | **OR** | | | | |
| 5. c) | How many integral solutions are there to $x_1 + x_2 + x_3 + x_4 + x_5 = 20$ where $x_1 >= -3$, $x_2 >= 0$, $x_3 >= 4$, $x_4 >= 2$, and $x_5 >= 2$? | 5 | 3 | 4 | 3 |
| 5. d) | In a survey of students at Florida State University the following information was obtained, 260 were taking a statistics course, 208 were taking a mathematics course, 160 were taking a computer programming course, 76 were taking statistics and mathematics, 48 were taking statistics and computer programming, 62 were taking mathematics courses and computer programming, 30 were taking all 3 kinds of courses, and 150 were taking none of the 3 courses.<br>Let<br>S = {students taking statistics},<br>M = {students taking mathematics},<br>C = {students taking computer programming}.<br>(i) How many students were surveyed?<br>(ii) How many students were taking a statistics and a mathematics course but not a computer programming course?<br>(iii) How many were taking a statistics and a computer course but not a mathematics course?<br>(iv) How many were taking a mathematics course but not taking a Statistics course or a computer programming course?<br>(v) How many were taking a computer programming course but not taking a course in mathematics or in statistics. | 5 | 2 | 4 | 2 |
| 6. a) | Verify whether the given graphs are Isomorphic or not?<br>  | 5 | 3 | 5 | 3 |
| 6. b) | Explain how Euler graphs are different from Hamiltonian Graphs. | 5 | 2 | 5 | 1 |
| | **OR** | | | | |
| 6. c) | Find the minimum spanning tree for the given graph using Prims's algorithm.<br>  | 5 | 3 | 5 | 1 |
| 6. d) | Define Chromatic Number. Explain the steps involved in graph coloring problem with suitable example. | 5 | 3 | 5 | 2 |

M: Marks; L: Bloom's Taxonomy Level; CO: Course Outcome; PO: Programme Outcome